Program:

Function code:

```
function root1(G, box, text)
evans(G);
xgrid();
a=gca();
if box~=0 then
a.box="on";
a.data_bounds=box;
end
a.children(1).visible='off';//remove the legend block
xtitle(text);
endfunction//save the code as root1.sce
```

Main program:

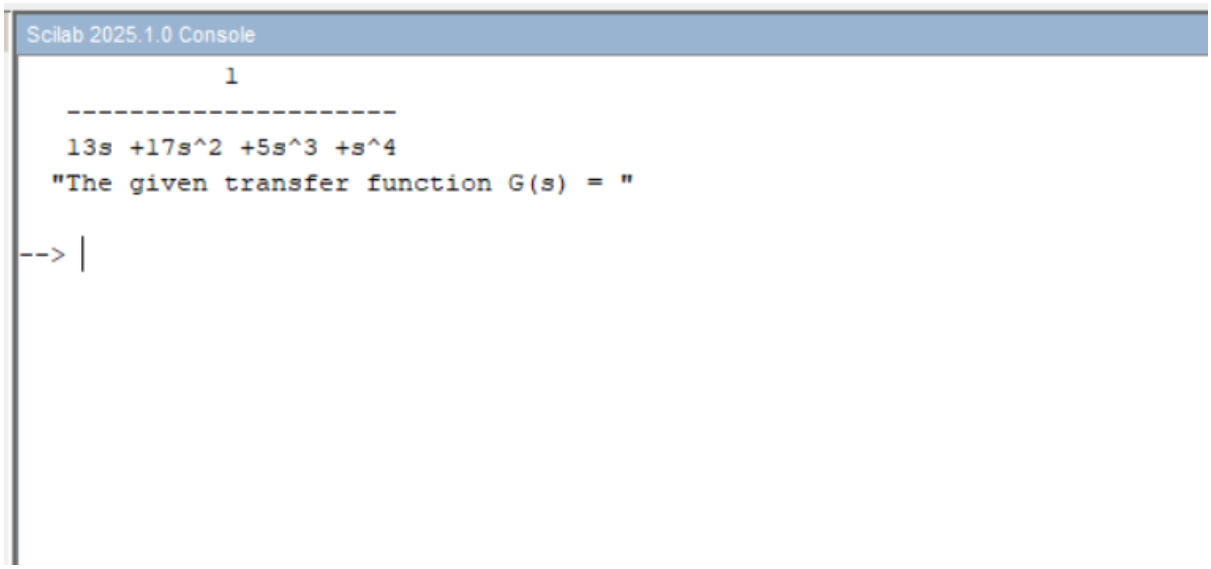
```
//Root locus
clear;
clc;
xdel(winsid());
//cd"your directory"
cd"C:\Users\admin\Documents"
exec("root1.sce");

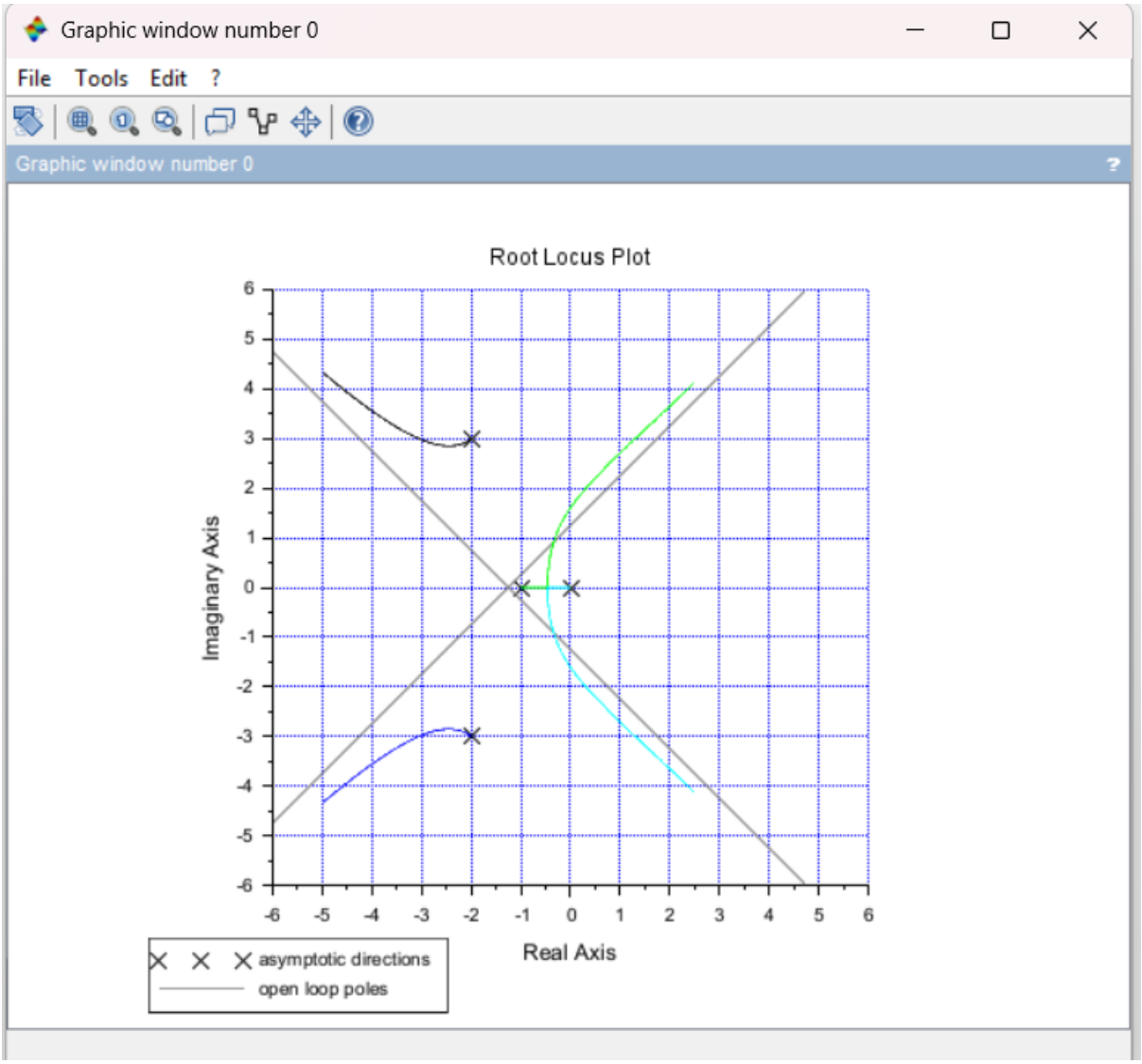
s=%s;
G=syslin('c',1,s*(s+1)*(s^2+4*s+13));
disp(G, 'The given transfer function G(s)=');
root1(G,[-6 -5;6 5],'Root locus plot for the given transfer function
G(s)=1/s(s+1)(s^2+4s+13)');
```

//simply write the transfer function and choose

//suitable range [xmin ymin:xmax ymax]

OUTPUT:




Graphic window number 0
File Tools Edit ?
Graphic window number 0
Root Locus Plot
Imaginary Axis
Real Axis
6
5
4
3
2
1
0
-1
-2
-3
-4
-5
-6
-6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6
asymptotic directions
open loop poles